🧾 Supervised Learning – Classification

🔎 What is Supervised Learning?

In **Supervised Learning**, a model learns from **labeled data**.
The data contains:

- 📝 **Input features (X)** → Information used for prediction.
- 🎯 **Output labels (Y)** → The correct answer for each input.

The model's job is to **learn a mapping function** from inputs to outputs.

Example:
✉️ Email text (input) → Label: **Spam** or **Not Spam** (output).

🏷️ What is Classification?

Classification is a Supervised Learning task where:

- The **output (Y)** is **categorical** (discrete classes or categories).
- The model assigns a **class label** to each input.

Simple Example:
🏠 Given details of a house, predict if its price category is **Low**, **Medium**, or **High**.

💧 Key Terminology

Term	Meaning
Features (X)	Variables or attributes used for prediction.
Label (Y)	Known category/class for each input.
Training Data	Data used to teach the model.
Testing Data	New data to evaluate model performance.
Classifier	The algorithm/model performing the classification task.

⚙️ How Classification Works – Workflow

1. 📥 **Collect labeled data**.
2. 🔎 **Choose a classification algorithm** (e.g., Decision Tree, K-Nearest Neighbors).
3. 🏋️ **Train the model** on training data.
4. 🧪 **Test the model** on unseen data.
5. 📊 **Evaluate performance** using metrics like Accuracy, Precision, Recall.

🧩 Common Classification Algorithms

Algorithm	Suitable For
Decision Trees	Easy to interpret problems.
K-Nearest Neighbors (KNN)	Simple pattern recognition tasks.
Naïve Bayes	Text classification, spam filtering.
Support Vector Machine (SVM)	Complex, high-dimensional data.
Logistic Regression	Binary classification problems.
Neural Networks	Large and complex classification tasks (e.g., image recognition).

📊 Example: Email Spam Classifier

Feature (X)	Label (Y)
Email contains words like "discount" or "offer"	Spam
Email contains official job title words	Not Spam

Task:
The model learns patterns from these features and classifies new emails as **Spam** or **Not Spam**.

📝 Evaluation Metrics (Key Performance Indicators)

Metric	Meaning
Accuracy	% of correct predictions out of all predictions.
Precision	% of correct positive predictions.
Recall	% of actual positives correctly predicted.
F1-Score	Balance between Precision and Recall (harmonic mean).

🎯 Real-World Examples

- 📧 **Spam Detection** → Classify emails as spam or not spam.
- 🏥 **Medical Diagnosis** → Classify patients as sick or healthy.
- 💳 **Banking** → Classify loan applications as approved or denied.
- 🖼️ **Face Recognition** → Identify people in photos.

🔍 Quick Summary

Task Type	Output	Example
Classification	Categories / Labels	Spam vs. Not Spam, Healthy vs. Sick